[GCT555 3DID]

ProxiWall: Distance-Driven Semantic Zoom for HMD-Free Virtual Museum Exploration on Wall Display

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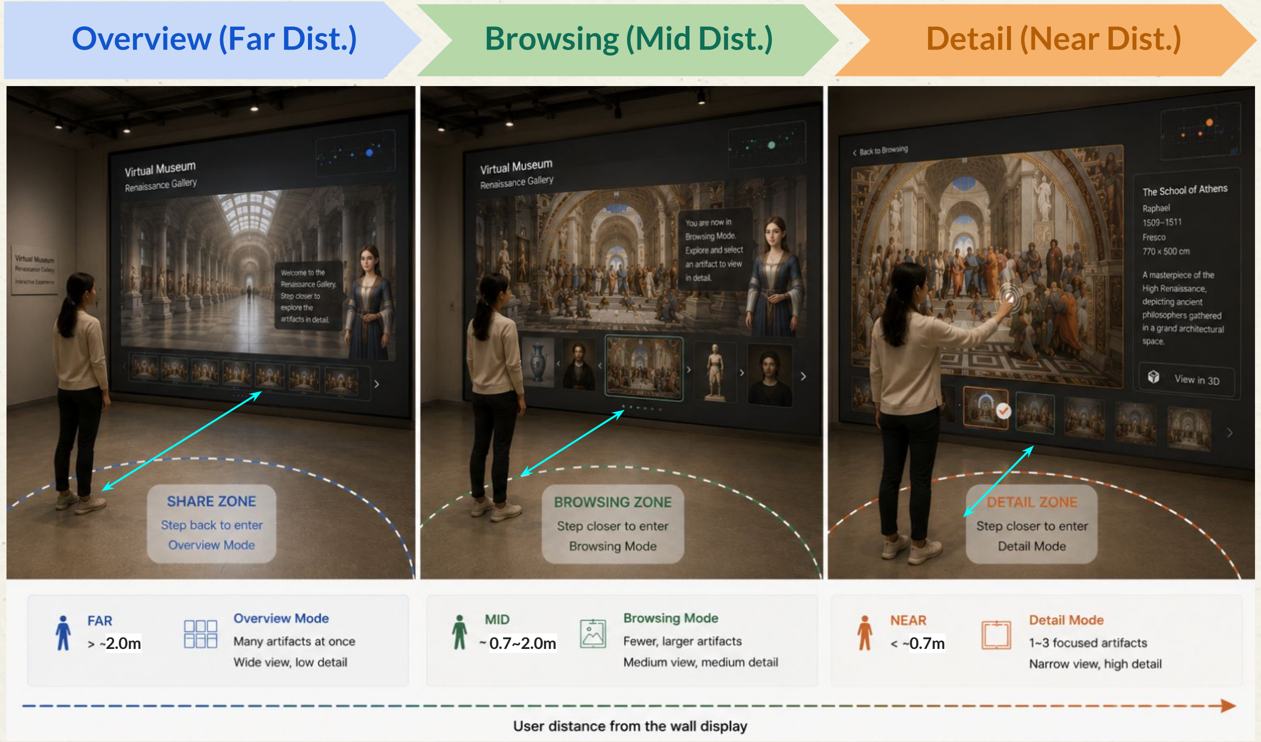
Distance-based Exploration

User distance from the wall automatically changes the view mode.



Research Question

Main RQ:

Are *distance-based body input* and *direct wall touch* meaningful interaction modalities for **the *Overview* → *Browsing* → *Selection* funnel** of a Wall Display virtual museum?

- OVERVIEW Stage (~2.0 m, FoV ≈ 37°, n=8-10)
- BROWSING Stage (0.7m~2.0 m, FoV ≈ 67°, n=4-6)
- SELECTION Stage (≤0.7 m + wall touch, n=1-3)

HeadZoom – 2025 IEEE ISMAR

- HeadZoom proposes **hands-free zooming and panning** using natural head motion.
- It maps **head translation** → **zoom** and **head yaw/pitch** → **image panning**.
- This is close to ProxiWall because ProxiWall also uses body movement as the main navigating interaction Input

Zhang, Kaining, et al. "HeadZoom: Hands-Free Zooming and Panning for 2D Image Navigation Using Head Motion." 2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR). IEEE, 2025.

Interaction Mapping

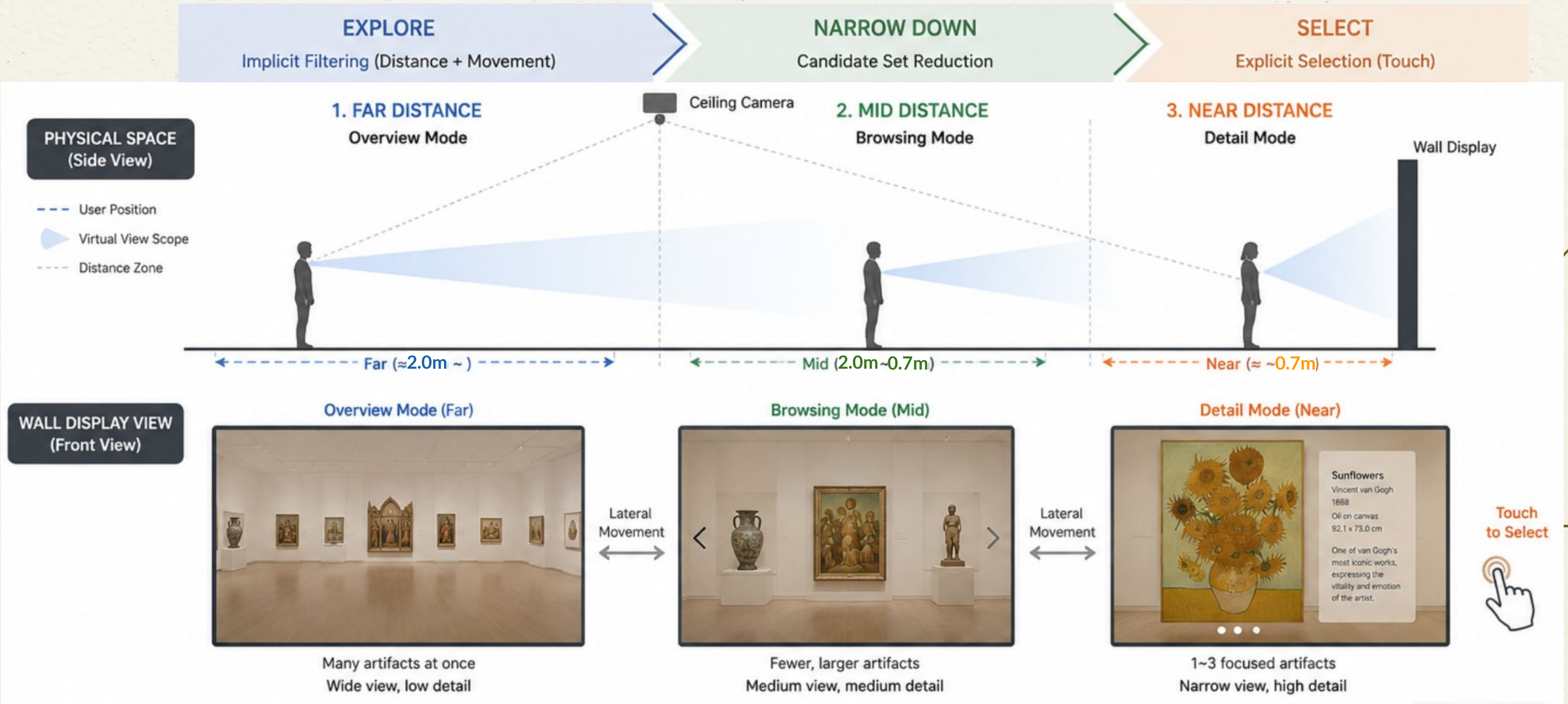
Interaction Element	HeadZoom	ProxiWall
Forward / backward movement	**Zoom in/out** by head distancer from image plane	Overview → Browsing → Detail by user-wall distance
Left / right movement	**Panning** target determined by head-direction raycasting	Gallery **panning** through lateral body movement
Input body part	Head position + head orientation	Whole-body position
Interaction type	Continuous zoom-and-pan control	Embodied museum exploration
Content	2D image / visual search	3D-rendered virtual museum / artifacts

Core Novelty

Beyond HeadZoom: Room-Scale Semantic Zoom on an LED Wall

- **Ceiling camera-based tracking**
  ProxiWall allows visitors to move naturally without wearing an HMD or holding a controller.
- **Larger-scale movement range**
  While HeadZoom uses small head motions, ProxiWall uses full-body walking distance and lateral movement within the exhibition space.
- **LED wall as a shared display**
  Instead of a private HMD view, ProxiWall presents the experience on a large wall display that can be seen by nearby visitors.
- **Semantic zoom, not just visual zoom**
  ProxiWall does not simply magnify the view. It changes the information level from **Overview** → **Browsing** → **Detail** based on the visitor's distance.

Interaction Framework



Overview

- **Environment setting**
  a. **Input** Multiple RGB-D Camera (Front view + Top view)
  b. **Output** 3D Rendered Video — LED Wall (w/o HMD)
- **Interaction feature**
  a. **Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
     i. Distance controls Overview → Browsing → Detail transition
  b. **Body-based Gallery Navigation**: Lateral movement controls horizontal exploration
  c. **Pseudo touch-based selection** : Touch confirms the final artifact selection

Interaction feature

- **Module**

1. **(z axis) Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
    a. Distance controls Overview → Browsing → Detail transition
2. **(x axis) Body-based Gallery Navigation**: Lateral movement controls horizontal exploration
3. **Pseudo touch-based selection** : Touch confirms the final artifact selection

Semantic Zoom

Module 01 · Z-axis

RATIONALE

CollabVisAdapt — Wang et al. (TVCG / IEEE VR 2026)

Uses user-to-object distance as a spatial context to auto-switch visualization modality. ProxiWall reuses this distance-as-trigger pattern for gallery semantic zoom.

Distance-Adaptive Visual Guidance — Choi et al. (ISMAR 2025)

Zone-Based Guide that switches modality across Far / Mid / Near zones — direct precedent for our Overview ⟶ Browsing ⟶ Detail tiers.

HeadZoom — Zhang et al. (ISMAR 2025)

Hands-free zoom driven by head translation. ProxiWall scales body-motion-as-zoom from micro head motion to full-body walking distance.

Hall (1966) — proxemic zones

Intimate / personal / social distance bands ground our 0.7 m and 2.0 m thresholds.

PARAMETERS

Far / Mid threshold

2.0 m

personal ↔ social boundary

Mid / Near threshold

0.7 m

personal ↔ intimate boundary

Hysteresis margin

± 0.15 m

prevents zone oscillation

Items per mode

8–10 / 4–6 / 1–3

Overview ⟶ Browsing ⟶ Detail

Gallery Navigation

Module 02 · X-axis

RATIONALE

Exploring Pointer — Bihani et al. (ACM ISS 2024)

On large curved displays, lateral offset from the central axis degrades pointing accuracy ≥ 6 % — motivates careful CD-gain tuning.

PerspectAR — Raza et al. (CHI 2025)

Perspective distortion on very large displays varies with lateral position. We adopt per-viewer mapping so the active candidate centres on each visitor.

Make Interaction Situated — Zeng et al. (CHI 2024)

Body movement is among the most user-acceptable inputs for situated public visualization — supports lateral walking over gestures or controllers.

Casiez & Vogel (CHI 2008)

Control-Display gain materially affects pointing speed and accuracy — basis for mode-dependent CD-gain.

PARAMETERS

Dead zone

± 10 cm

ignores postural sway

CD-gain · Overview

1 m → 1 column

broad, low-gain scan

CD-gain · Browsing

1 m → 1–2 items

candidate-level granularity

Max scroll velocity

≤ 2 items / s

prevents visual nausea

Pseudo Touch

Module 03 · Wall contact

RATIONALE

VisTorch — Patnaik, Peng & Elmqvist (CHI 2024)

Touch confirms the final target in a body-pointed situated-visualization workflow — mirrors our body-reveal → touch-confirm pattern.

Exploring Pointer — Bihani et al. (ACM ISS 2024)

Direct selection accuracy degrades with distance and lateral offset — justifies a near-zone touch step instead of long-range pointing.

Expert Experiences — Sinaei Hamed et al. (ACM ISS 2024)

Practitioners cite cursor visibility and interaction interruption as core large-display pain points — wall touch sidesteps both.

Ng et al. (UIST 2012) · Jota et al. (CHI 2013)

< 100 ms feels direct; < 50 ms is imperceptible — sets the touch-latency budget.

PARAMETERS

Touch target size

≥ 44 mm

Fitts minimum for direct selection

Touch latency budget

≤ 100 ms

below JND for direct manipulation

Confirmation rule

Single tap

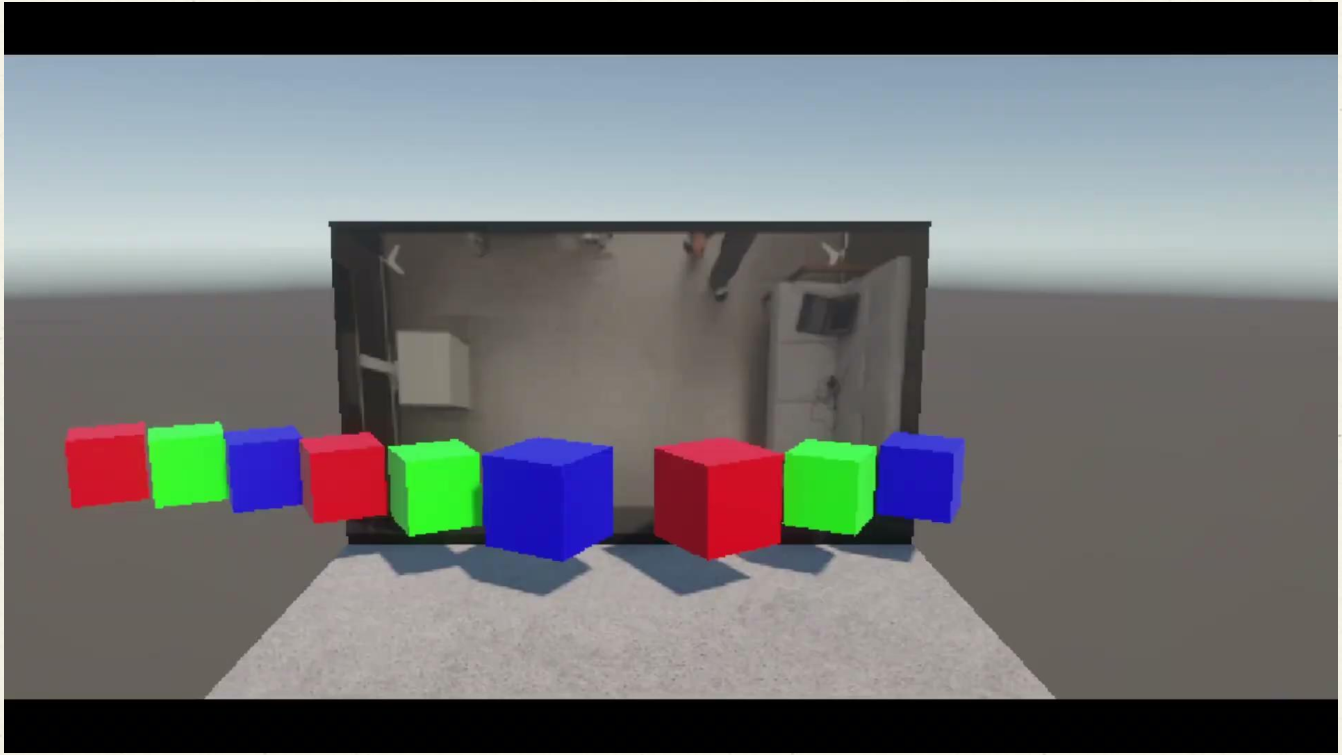
fast museum throughput

Feedback

Ripple + click

+ undo via step-back

Module 1&2



Module 1&2

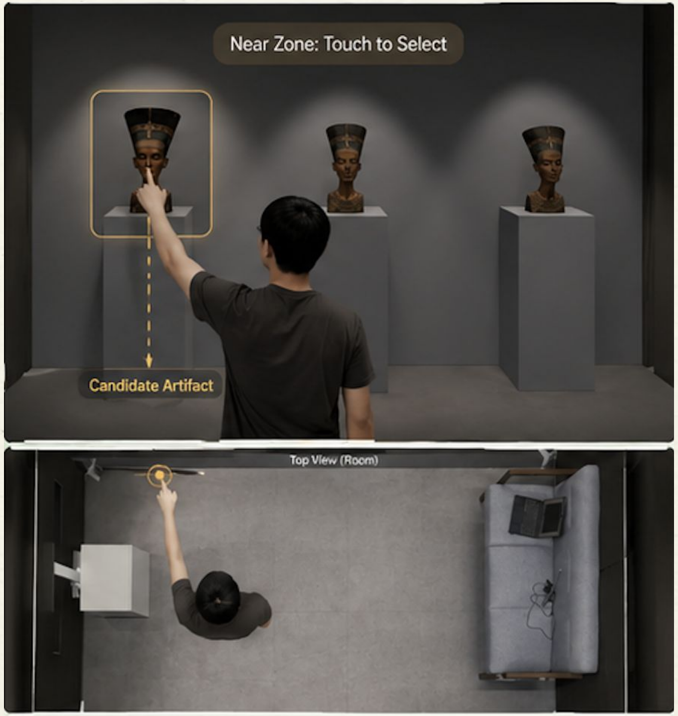
Calculate pose center by getting mean value of shoulders and hips

```
center.x = (leftShoulder.x + rightShoulder.x + leftHip.x + rightHip.x) * 0.25f;
center.y = (leftShoulder.y + rightShoulder.y + leftHip.y + rightHip.y) * 0.25f;
```

We treat this as a torso midpoint which is the user’s position in the camera image.

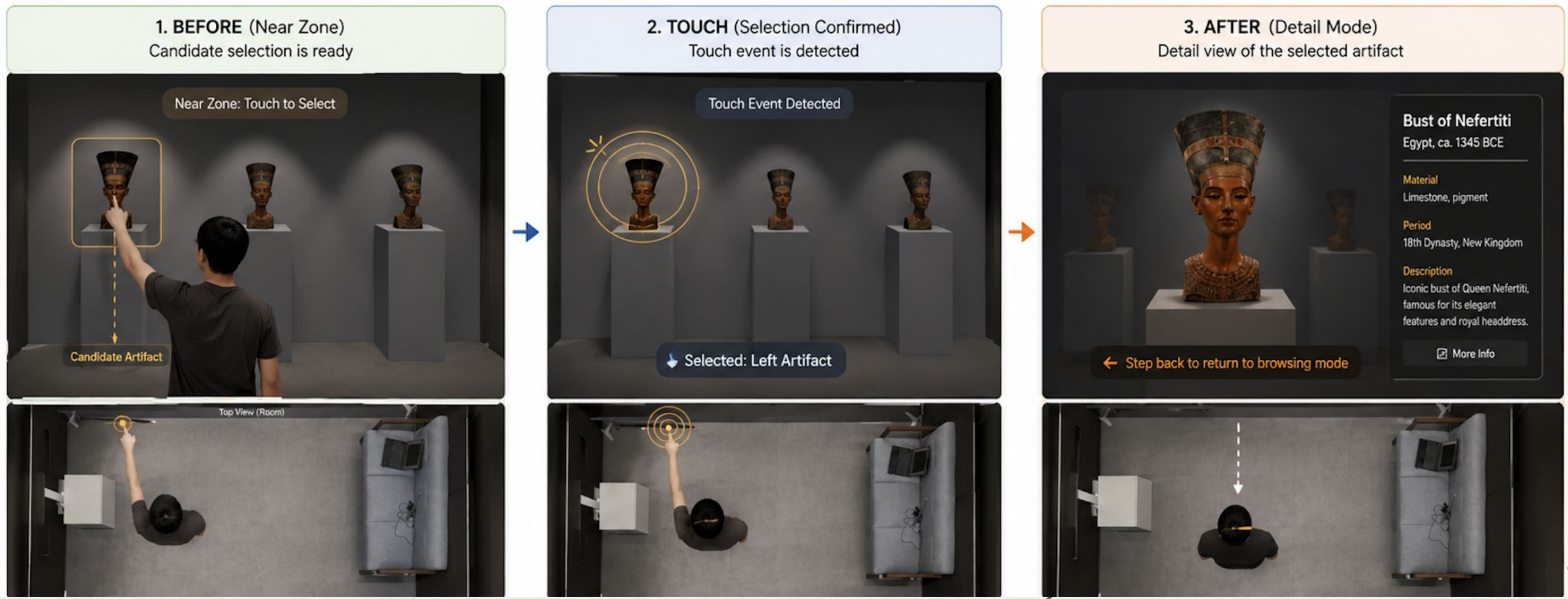
Then, we decide where the artifacts should be placed in the 3D space.

Module 3



Environment Setting

Touch based Part Selection : From candidate to detail view

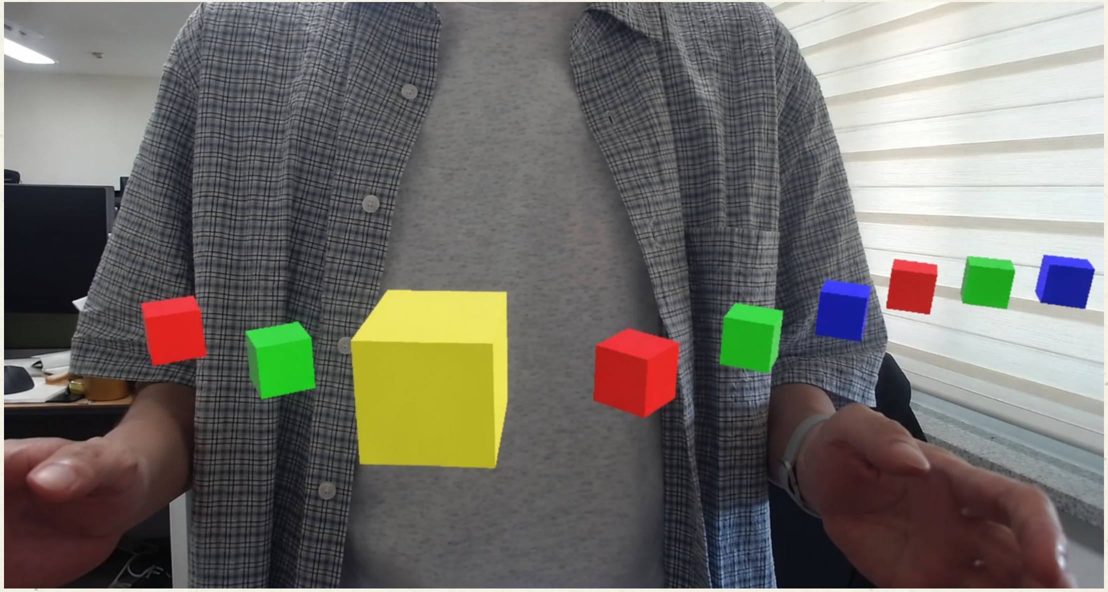


Environment Setting

Virtual Museum (Select Mode : Obj n=3)

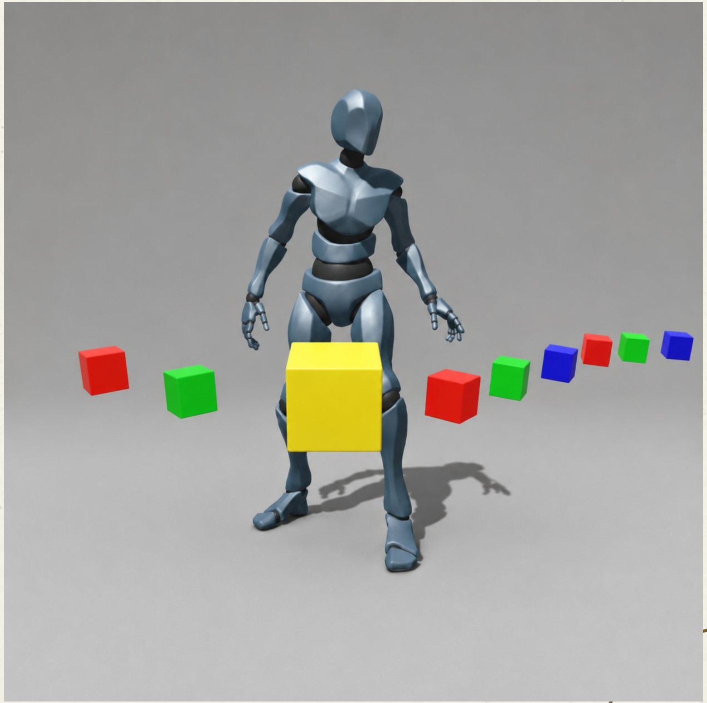
Environment Setting

Front Cam Problem

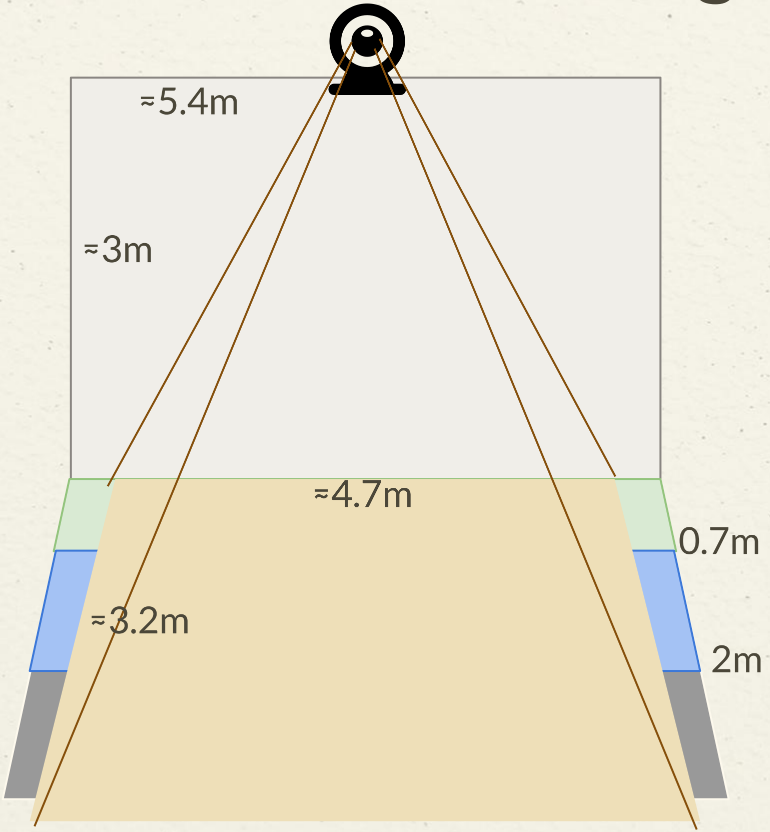


Environment Setting

Changed to avatar controlled by Ceiling Cam

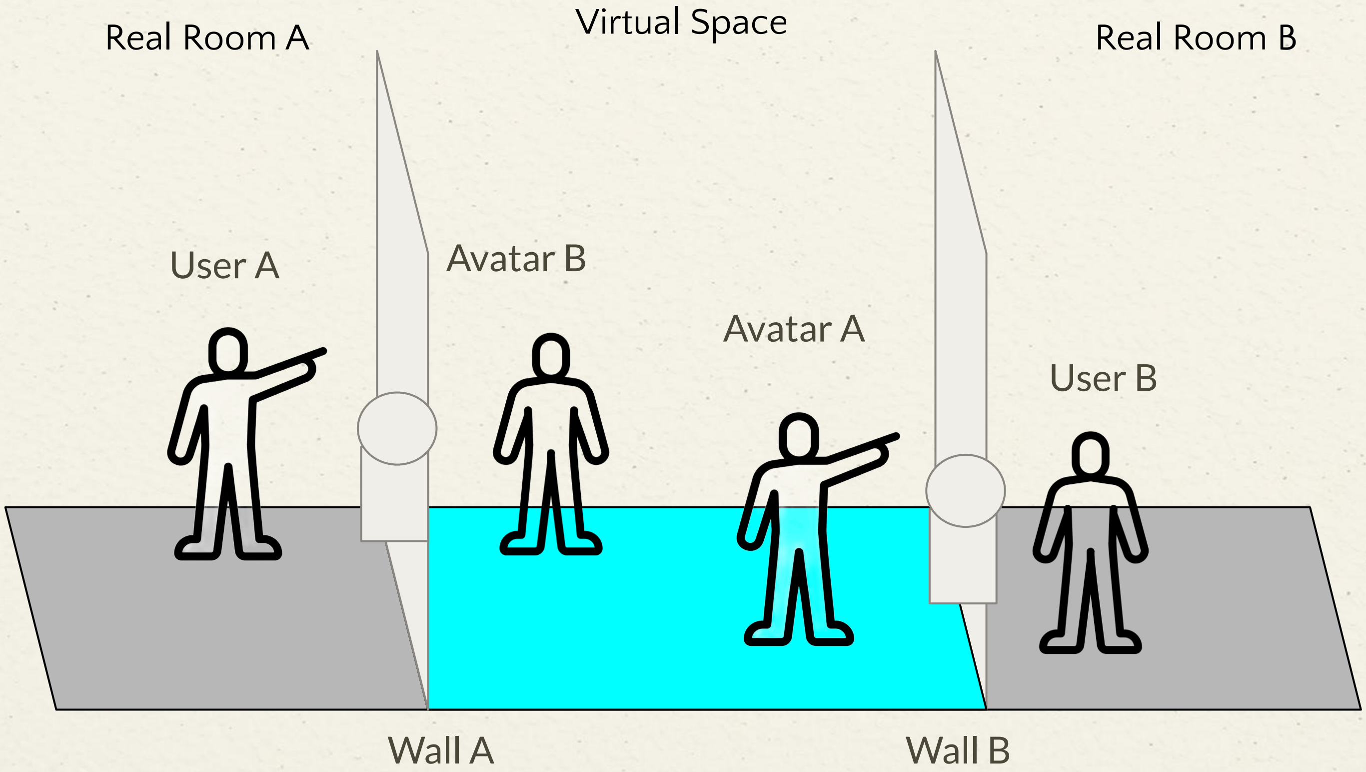


Environment Setting

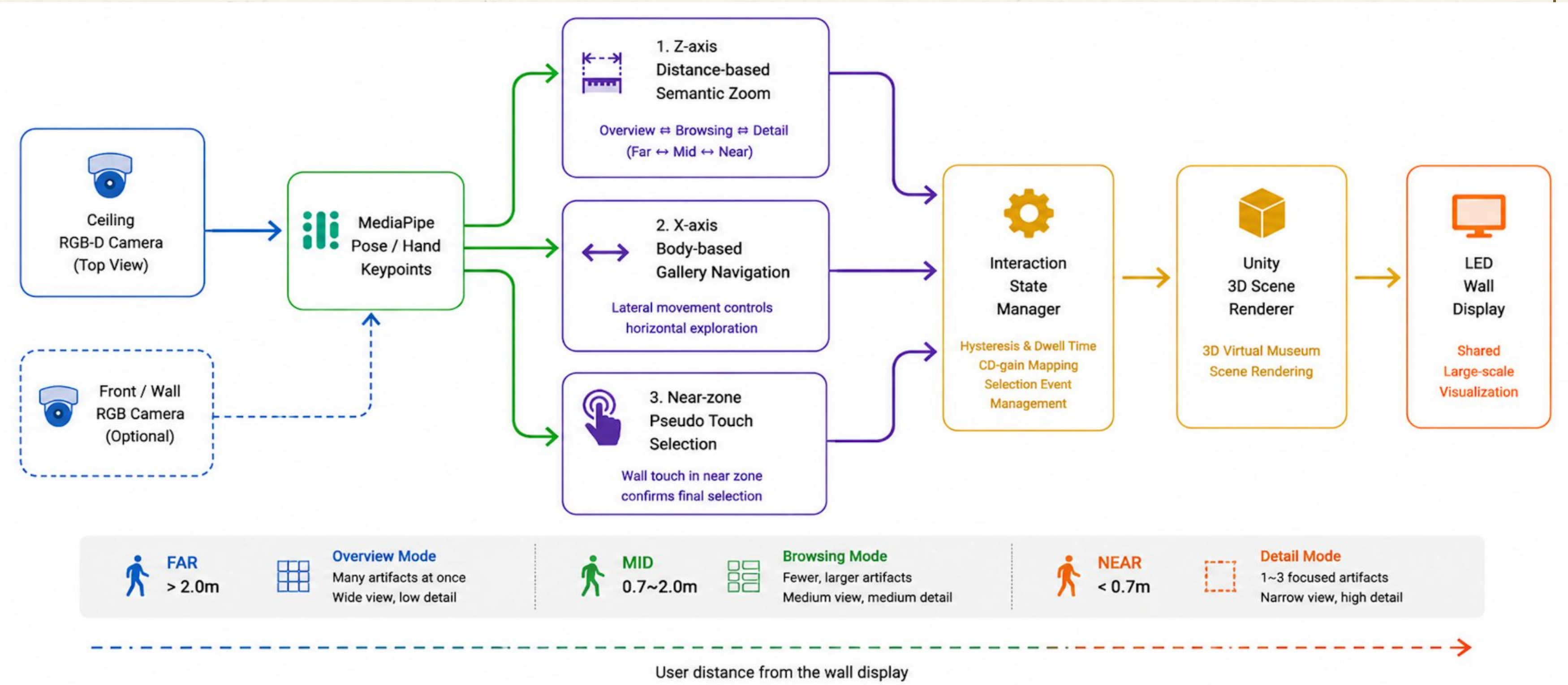


= Overview Mode
= Browsing Mode
= Detail Mode
= LED Wall
= Captured Area

Environment Setting–Dual Wall



Implementation



Development Road Map

Done

9	11	12	13	14	15
Sensing & Tracking	**Distance & Hierarchy**	**Transition Logic**	**Lateral & Touch**	**Multi-Visitor**	**Evaluation**
Ceiling RGB-D Camera	Zone thresholds + Overview/Browsing/Detail layouts	Hysteresis, dwell, animation curves	CD-gain tuning + pseudo-touch frame	Shared/private switch, conflict resolution	Study Design -> Development for Experiment -> Analysis

Timeline

Description	Sub-task	Week 7	8	9	10	11	12	13	14	15	16	Owner	Status
Interaction Concept	Ideation + scenarios											HJ	Completed
Tracking Setup	Ceiling RGB-D rig install											SJ	Completed
	Skeleton + pipeline											JJ	Completed
Unity Prototype	Scene + asset import											JJ	Completed
	Wall display renderer											JJ	Completed
	Sensor → Unity bridge											JJ	Completed
Parameter Design	Distance zone calibration											HJ	In Progress
	Hysteresis & dwell tuning											HJ/SJ	In Progress
	CD-gain function design											SJ	In Progress
	Touch size & latency spec											HJ	In Progress
Semantic Zoom	Overview mode layout											SJ	In Progress
	Browsing + focus indicator											SJ	In Progress
	Detail + metadata panel											SJ	In Progress
FoV Mapping	Wall geometry calibration											JJ	Not started
	Angular mapping function											JJ	Not started
	Multi-user perspective											JJ	Not started
User Study	Pilot (n = 3–5)											SJ	Not started
	Main study (n = 15–20)											SJ	Not started
Analysis	Quantitative analysis											HJ	Not started
	Qualitative analysis											HJ	Not started
	Writing & figure prep											HJ	Not started

OWNERS HJ · Heejeong SJ · Seungjae JJ · Jungjin

Thanks!

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Problem & Need : UX Scenario

- General museum visitors need an accessible, shared, and intuitive way to explore virtual museum spaces without wearing HMDs or using controllers.
- **Problem:** A fixed wall display has only one viewpoint — yet far-viewers and near-viewers each want a different one. Without an HMD, today's wall systems cannot serve both.

FAR · Overview — visitors scan the room, taking in many works at a glance.

NEAR · Detail — visitors stp, focus, and examine a single work.

▸ **[The core idea]:** Museum visitors should be able to explore virtual galleries the way they explore real ones — by **walking and looking** — without strapping on a headset.

Approach

- **Environment setting**
  a. Input Multiple RGB-D Camera (Front view + Top view)
  b. Output 3D Rendered Video — LED Wall (w/o HMD)
- **Interaction feature**
  a. **Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
     i. Distance controls Overview → Browsing → Detail transition
  b. **Body-based Gallery Navigation**: Lateral movement controls horizontal exploration
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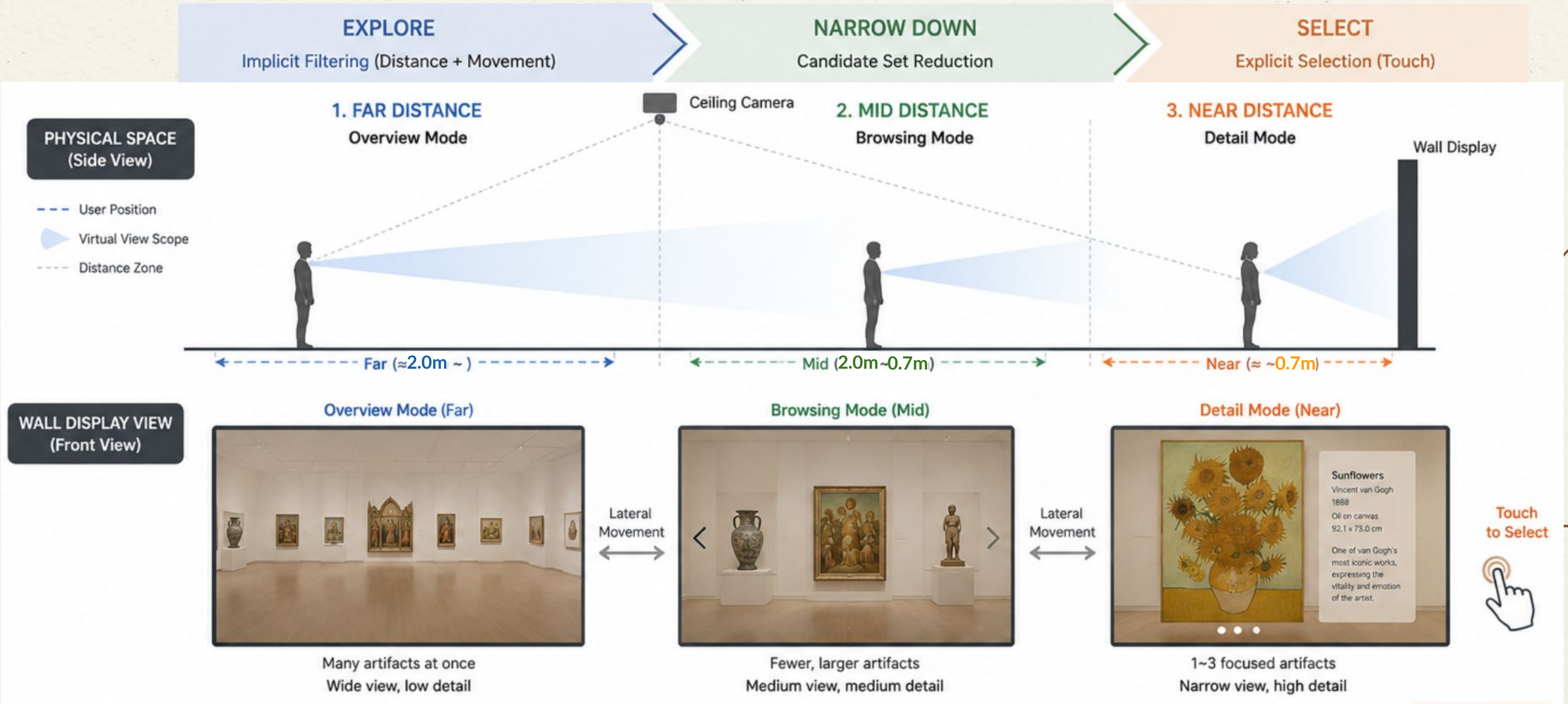
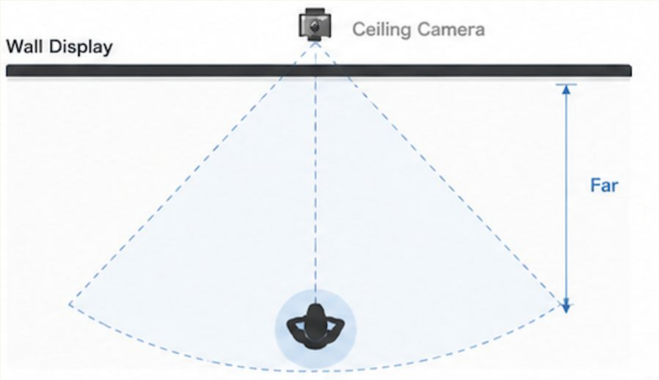
Exploration Mode (Overview, Browsing, Detail)



[Far] Overview mode



Overview

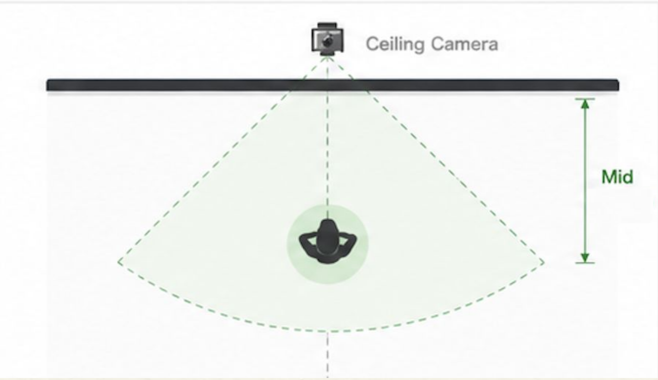
Peripheral + UFOV scanning · n bounded by Miller 7±2 (loose) · each obj at ~5–7° angular size for floor recognizability Greenberg et al.'s Proxemic Interactions *(2011)* ; Miller 1956 ; Vogel & Balakrishnan 2004]

1	2	3
Overview	**Browsing**	**Detail**
~2.0 m	0.7m~2.0 m	≤0.7 m + wall touch
FoV 53°	FoV 53~110°	FoV 110°
n=8–10	n=4–6	n=1–3

Distance (m)

Wall Display

[Mid] Browsing mode



Browsing

Dynamic foveal-parafoveal compare zone · n reduced per Hick-Hyman $\log_2(n+1)$ · fits 1–2 saccade cycle [Miller 1956 ; Hick 1952 ; Hyman 1953]

1

Overview

~2.0 m
FoV 53°
n=8–10

2

Browsing

0.7m~2.0 m
FoV 53~110°
n=4–6

3

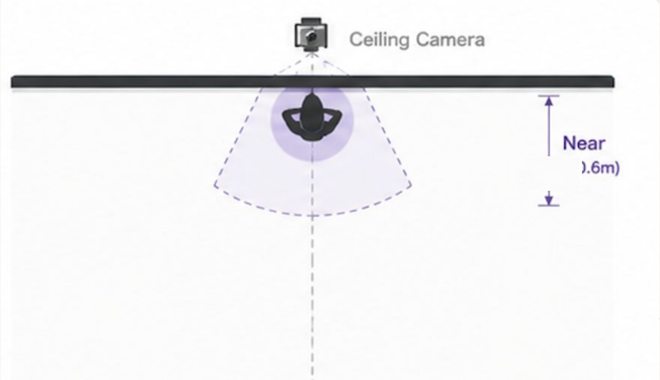
Detail

≤0.7 m + wall touch
FoV 110°
n=1–3

Wall Display

Distance (m)

[Near] Detail mode



Detail

1 obj subtends > 50° → foveal / ventral identification · Hick-Hyman RT ≈ 0 at n = 1–3 · Fitts target W maximized for explicit commit [Goodale & Milner 1992 ; Fitts 1954 ; Dourish 2001]

1	2	3
Overview	**Browsing**	**Detail**
~2.0 m	0.7m~2.0 m	≤0.7 m + wall touch
FoV 53°	FoV 53~110°	FoV 110°
n=8–10	n=4–6	n=1–3

Distance (m)

Wall Display

Comparisons

	ProxiWall	CollabVisAdapt (2026/VR)	Blanc	Co-Living between Reality & Virtuality (2026/CHI)
Main Focus	Collaborative 3D exploration	Distance-adaptive MR visualization	Vision-based wall touch	HMD-free virtual–physical integration
Interaction	Gesture + pseudo touch + sound	Distance-based adaptation	Fingertip touch detection	Natural body/touch interaction
Core Idea	Shared ↔ Private switching	Visualization switching	Surface touch sensing	Reality–virtuality coexistence
Key Difference	Multimodal collaborative workflow	Rendering adaptation only	Touch input only	Conceptual framework
Display	Dual LED wall	HMD AR/VR	HMD + RGB camera	Spatial / wall displays

Hypotheses

H1 The proposed interaction will result in significantly shorter **Task Completion Time** compared to the baseline *(Controller-based Manipulation)*.

H2 The proposed interaction will result in significantly higher **Selection Accuracy** compared to the baseline.

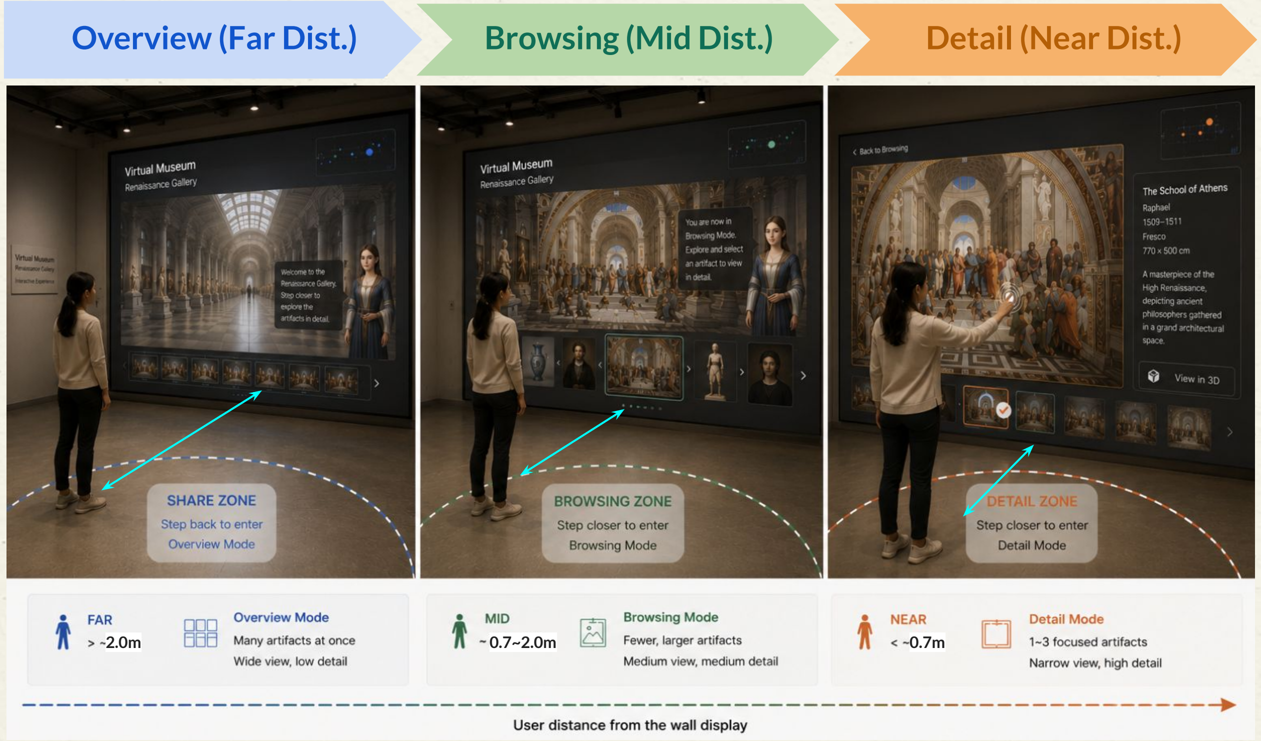
H3 The proposed interaction will result in significantly lower **Selection Error Rate** compared to the baseline.

H4 The proposed interaction will improve **Exploration Efficiency**, as reflected by fewer non-target steps before reaching the correct target compared to the baseline.

H5 The proposed interaction will result in lower perceived cognitive workload, as measured by **NASA-TLX** and higher **Spatial Presence** as measured by MEC-SPEQ compared to the baseline.

Distance-based Exploration

User distance from the wall automatically changes the view mode.



Evaluation

Objective Measures

Metric	Measurement
Task Completion Time	Time from Task Start to Target Artifact Selection (seconds)
Exploration Efficiency	Number of non-target artifacts explored or interaction steps before correct target selection.
Selection Accuracy	Percentage of trials in which the participant correctly selected the target artifact.
Selection Error Rate	Percentage of trials involving incorrect object selection or failure to select the target artifact.

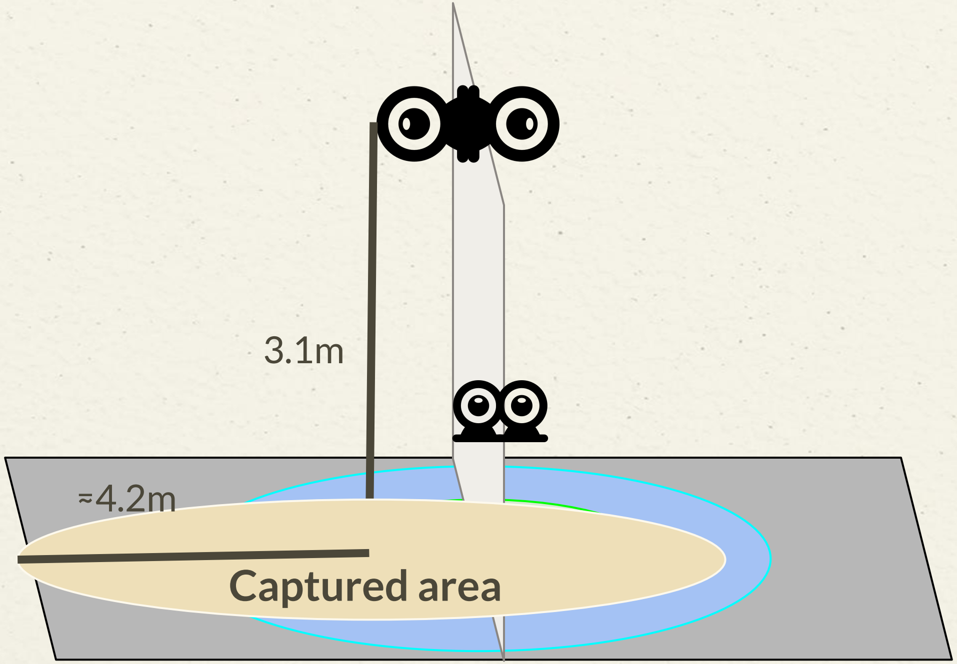
Subjective Measures

Metric	Measurement
NASA-TLX	Workload, mental demand, and physical demand
Presence (MEC-SPQ, 8 items)	Spatial situation model, self-location, and possible actions in the virtual museum

Qualitative Evaluation

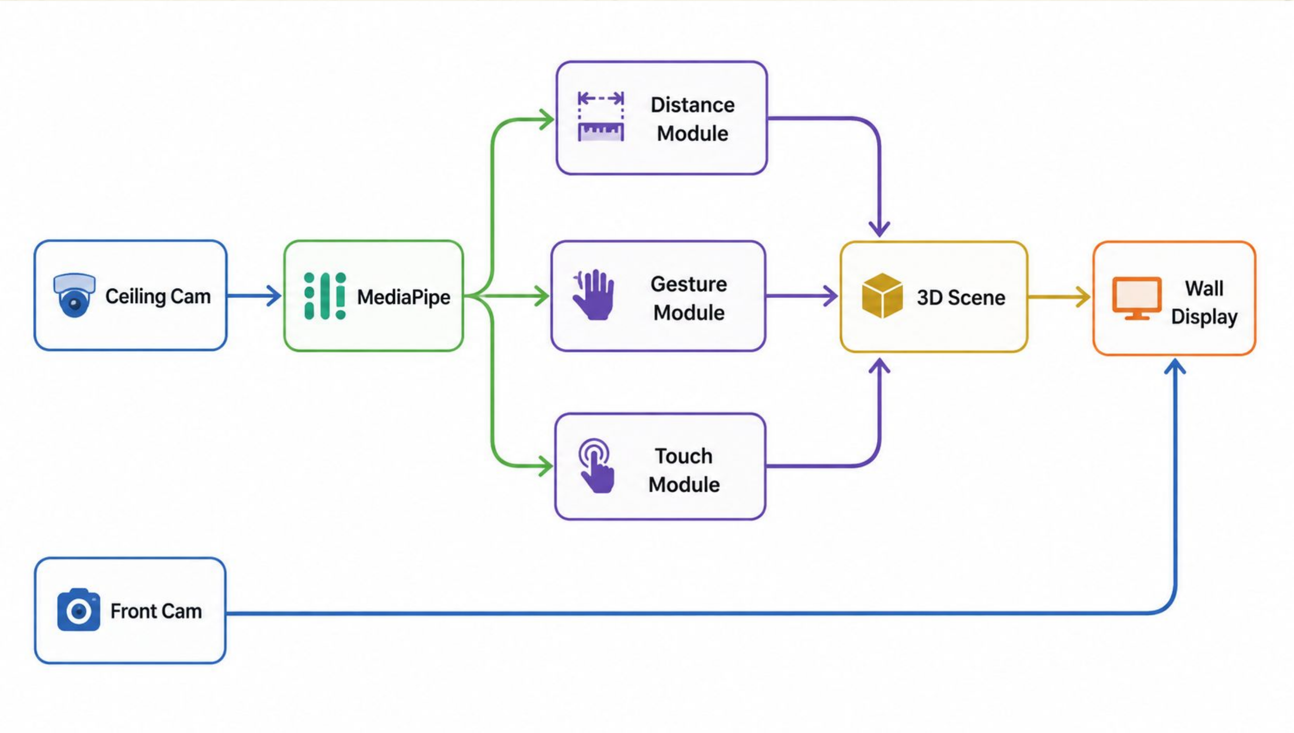
Semi-Structured Interview	Naturalness, immersion, discomfort, and **preference**

System Setup

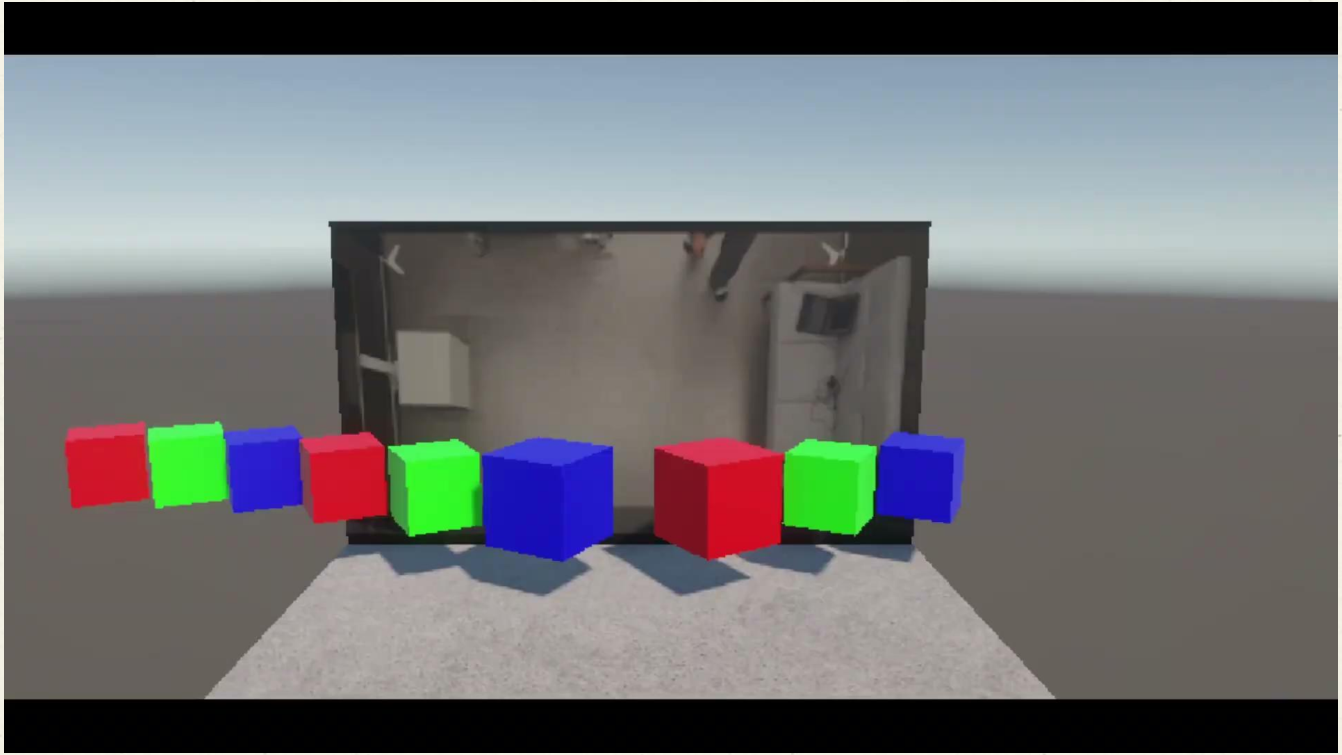


= Overview Mode
= Browsing Mode
= Detail Mode
= LED Wall
14mm crop body camera

Implementation



Demo.



Timeline

Description	Week										Status
	7	8	9	10	11	12	13	14	15	16	
Interaction Concept											Completed
Tracking Setup											Completed
Unity Prototype											Completed
Parameter Design for Interaction											In Progress
Distance-Based Semantic Zoom											In Progress
FoV-Based Object Mapping											Not started
User Study											Not started
Analysis											Not started

Heejeong Seungjae Jungjin